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vÊfz\ n8b T -Æ^sSð|û~ß V3.2                                {, 1 ~u
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```

```
===== js/app.js =====
```

```
/**
 * vÊfz\ n8b T -Æ - App Shell
 * @author ch2077
 * @version 3.2.0
 * @date 2025-05-10 etT 21k>n8b
 */
n8b hFgŷ j!WWS gŷg„ÿ kÎN*n8b rizËjsj!WW
~ßN u T}Thg ŷ init(container, statsCb) !' destroy()
~ß<¡VÐÆ ŷ updateStats({key:value}) R7e~v•ërŷ` h
*/
const app = {}
  currentGame: null,
  games: [
    {
      id: 'memory',
      name: '°_Æ•ûrL',
      icon: '0>Yà',
      desc: '•û_ SarGb~R0v0T v„'M[ûÿ € ŠEO`v„<°_ÆR>ÿ ',
      tags: ['°_Æ', 'M[û'],
      module: MemoryGame
    },
    {
      id: 'snake',
      name: '•*T †Ç',
      icon: '0=Û ',
      desc: '~ÏQx•*T †Çn8b ŷ T c%~ßri•Š••ŠY'ÿ ',
      tags: ['~ÏQx', '^Wg:'],
      module: SnakeGame
    },
    {
      id: '2048',
      name: '2048',
      icon: '0=Y"',
      desc: 'nÑR``T ^vep[We¹WWÿ c b •%R0 2048ÿ ',
      tags: ['ep[W', '{Vue'],
      module: Game2048
    }
  ],
```

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```
{ð
  id: 'minesweeper',ð
  name: 'bk÷',ð
  icon: 'ø=Ü£',ð
  desc: '~İQxbk÷n8b ŷ ••_ wø÷b~Qúb@g [%Qhh<ŷ ',ð
  tags: ['-İQx', 'c`'t '],ð
```

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=====

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=====

```
    module: MinesweeperGameð
  },ð
  {ð
    id: 'puzzle',ð
    name: 'nÑWwbüVp',ð
    icon: 'ø>Ÿé',ð
    desc: 'yÜR`nÑWw•øSŸVprG~z^•ŷ ~İQxv,, 15 büVpŷ ',ð
    tags: ['•;•'', 'c'^^•'],ð
    module: PuzzleGameð
  },ð
  {ð
    id: 'tetris',ð
    name: 'OÄ•We e¹WW',ð
    icon: 'ø>Ÿñ',ð
    desc: '~İQxOÄ•We e¹WWŷ m^-det^L_-R SG~Sŷ ',ð
    tags: ['-İQx', '^Wg:'],ð
    module: TetrisGameð
  },ð
  {ð
    id: 'tictactoe',ð
    name: 'N•[WhÉ',ð
    icon: '+U',ð
    desc: '~İQxN [PhÉŷ N AI [üb w Æ QH•Pb N ~¿ŷ ',ð
    tags: ['{Vue', '[üb '],ð
    module: TicTacToeGameð
  },ð
  {ð
    id: 'numberguess',ð
    name: 'ep[ws Æ ',ð
    icon: 'ø<ß',ð
    desc: 's Qú 1~100 NK•ôv,,y^yøep[wŷ f Vô0 • n •)\ ŷ ',ð
    tags: ['•;•'', 'c`'t '],ð
    module: NumberGuessGameð
  },ð
  {ð
    id: 'breakout',ð
```

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```
name: 'bSx wW',ð
icon: '0<ß0',ð
desc: '~İQxbSx wWÿ yûR`c!g•Sİ_9\ t m^-db@g x wWÿ ',ð
tags: ['-İQx', '^wg:'],ð
module: BreakoutGameð
},ð
{ð
  id: 'reaction',ð
  name: 'Sİ^"mK<0',ð
  icon: '&i',ð
  desc: 'mK<00`v„Sİ^"• ^!ÿ w R0-ÿ,rzËSsp¹Qûÿ ',ð
  tags: ['Sİ^"', 'c b '],ð
  module: ReactionGameð
},ð
```

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{, 3 ~u

```
{ð
  id: 'sudoku',ð
  name: 'epri',ð
  icon: '0=ÿ"',ð
  desc: '~İQxN][«h<ep[Wc` t ÿ Xknáb@g zzh<ÿ ',ð
  tags: ['ep[W', 'c` t '],ð
  module: SudokuGameð
},ð
{ð
  id: 'flappy',ð
  name: '\ ž ~p~p',ð
  icon: '0=0$',ð
  desc: 'p¹Qû<@ \ ž ~p•wÿ z••Ç{i•Sÿ •Š•Û•ŠY}ÿ ',ð
  tags: ['0 •ò', '^wg:'],ð
  module: FlappyGameð
},ð
{ð
  id: 'fruitcatch',ð
  name: 'c¥l4gæ',ð
  icon: '0<ßN',ð
  desc: 'nÑR`{i[Pc¥00l4gæÿ \ _Ã••_ p,_9ÿ ',ð
  tags: ['0 •ò', 'Sİ^"',ð
  module: FruitCatchGameð
},ð
{ð
  id: 'colormatch',ð
  name: '~æ,rS9'M',ð
  icon: '0<ß`',ð
```

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```
desc: ' _ë• R$e-e‡[W~æ,rf/T&N T yðN •ôÿ 30y0c b ÿ ',ð
tags: ['Sİ^"', '• R>'],ð
module: ColorMatchGameð
},ð
{ð
  id: 'delta',ð
  name: 'N %0m2'M^A',ð
  icon: '0<ß-p ',ð
  desc: 'N %0m2^LR"-•g:'M^ÄVhÿ w0Vp0 ^rTX0 g^h°Qh-•g:ÿ QÆY QúQûÿ ',ð
  tags: ['-•g:', 'M^Ä'],ð
  module: DeltaGameð
},ð
{ð
  id: 'bubble',ð
  name: 'lálá\ bk',ð
  icon: '0>þç',ð
  desc: '~İQxláláÿ™ÿ w,,QÆ\ Qûÿ m^-d3N*T ,rláláÿ ',ð
  tags: ['-İQx', 'Flash'],ð
  module: BubbleGameð
},ð
{ð
  id: 'invaders',ð
```

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{, 4 ~u

```
name: 'Y*zz0µue€ ',ð
icon: '0=Û~',ð
desc: '~İQx^Wg:\ Qûÿ m^pmY f ,0- ÿ OÝSkw0t ÿ ',ð
tags: ['^Wg:', '\ Qû'],ð
module: InvadersGameð
},ð
{ð
  id: 'wordle',ð
  name: 's SU<İ',ð
  icon: '0=Ÿ$',ð
  desc: '6k!g:0 s Qú5[WkÍ,ñe‡SU<İÿ [WkÍS0,rcðy:ÿ ',ð
  tags: ['SU<İ', 'c`t '],ð
  module: WordleGameð
},ð
{ð
  id: 'simon',ð
  name: '°_Æpo',ð
  icon: '0=Ûi',ð
  desc: '°00poQI~z^•^v'ÍY ÿ •Šge•Š••c b g•-Pÿ ',ð
  tags: ['°_Æ', 'c b '],ð
```

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```
    module: SimonGameĐ
  },Đ
  {Đ
    id: 'whack',Đ
    name: 'bSw0Ÿ ',Đ
    icon: '0=Ü9',Đ
    desc: '30y0-PeöbSw0Ÿ Ÿ w<u%bK_ë•ŠY •ŠY}Ÿ ',Đ
    tags: ['Sİ^"', '0 •ò'],Đ
    module: WhackGameĐ
  },Đ
  {Đ
    id: 'match3',Đ
    name: '[*wóm^m^NP',Đ
    icon: '0=ÜŽ',Đ
    desc: 'Næcbv0•»[*wóŸ 3N*NÂN •b~¿m^-dŸ •b• •bQûŸ ',Đ
    tags: ['m^-d', '{Vue'}],Đ
    module: Match3GameĐ
  }Đ
],Đ
Đ
init() {Đ
  this.renderGameGrid();Đ
  this.bindEvents();Đ
},Đ
Đ
renderGameGrid() {Đ
  const grid = document.getElementById('gameGrid');Đ
  grid.innerHTML = this.games.map(g => `Đ
    <div class="game-card" data-game="${g.id}">Đ
      <div class="game-icon">${g.icon}</div>Đ
```

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```
    <div class="game-name">${g.name}</div>Đ
    <div class="game-desc">${g.desc}</div>Đ
    <div class="game-tags">${g.tags.map(t => `<span class="tag">${t}</span>`
pan>`).join('')}</div>Đ
  </div>Đ
  `).join('');Đ
},Đ
Đ
bindEvents() {Đ
  document.getElementById('gameGrid').addEventListener('click', (e) => {Đ
    const card = e.target.closest('.game-card');Đ
    if (card) {Đ
```

```
const gameId = card.dataset.game;ð
this.openGame(gameId);ð
}ð
});ð
ð
document.getElementById('backBtn').addEventListener('click', () => {ð
  this.closeGame();ð
});ð
},ð
ð
openGame(gameId) {ð
  const gameDef = this.games.find(g => g.id === gameId);ð
  if (!gameDef) return;ð
ð
  this.currentGame = gameDef;ð
ð
  document.getElementById('homePage').style.display = 'none';ð
  document.getElementById('gamePage').style.display = 'block';ð
  document.getElementById('backBtn').style.display = 'block';ð
  document.getElementById('gameTitle').textContent = gameDef.name;ð
ð
  const controls = document.getElementById('gameControls');ð
  controls.innerHTML = `ð
    <div class="game-stats" id="gameStats"></div>ð
    <button class="btn small" id="gameRestart">'İe°_ YË</button>ð
  `;ð
  document.getElementById('gameRestart').addEventListener('click', () =>
{ð
  this.startGame();ð
});ð
ð
  document.getElementById('gameArea').innerHTML = '';ð
  this.startGame();ð
},ð
ð
closeGame() {ð
  if (this.currentGame && this.currentGame.module && this.currentGame.mod
ule.destroy) {ð
    this.currentGame.module.destroy();ð
  }ð
  this.currentGame = null;ð
}
```

```
document.getElementById('homePage').style.display = 'block';ð
```

```
document.getElementById('gamePage').style.display = 'none';Đ
document.getElementById('backBtn').style.display = 'none';Đ
document.getElementById('gameArea').innerHTML = '';Đ
},Đ
Đ
startGame() {Đ
  if (!this.currentGame) return;Đ
  const area = document.getElementById('gameArea');Đ
  area.innerHTML = '';Đ
  this.currentGame.module.init(area, this.updateStats.bind(this));Đ
},Đ
Đ
updateStats(stats) {Đ
  const el = document.getElementById('gameStats');Đ
  if (el) {Đ
    el.innerHTML = Object.entries(stats).map(([k, v]) =>Đ
      `<div class="stat">${k}: <span>${v}</span></div>`Đ
    ).join('');Đ
  }Đ
}Đ
};Đ
Đ
document.addEventListener('DOMContentLoaded', () => app.init());Đ

===== index.html =====

<!DOCTYPE html>Đ
<html lang="zh-CN">Đ
<head>Đ
  <meta charset="UTF-8">Đ
  <meta name="viewport" content="width=device-width, initial-scale=1.0">Đ
  <title>vĖfz\ n8b </title>Đ
  <link rel="stylesheet" href="css/style.css">Đ
</head>Đ
<body>Đ
  <div id="app">Đ
    <header class="app-header">Đ
      <div class="header-left">Đ
        <span class="logo">Đ<ß@</span>Đ
        <h1>vĖfz\ n8b </h1>Đ
      </div>Đ
      <button class="btn-back" id="backBtn" style="display:none">!• •ŦVÞ</button>Đ
    </header>Đ
  </div>Đ
  <main id="mainContent">Đ
    <!-- Home page - game grid -->Đ
    <div id="homePage" class="home-page">Đ
```

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```
<div class="hero">ð
  <h2>• bÉN N*n8b _ YËT'y </h2>ð
```

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```
    <p>0 •ðe>g-ÿ •;p¼` ~ô</p>ð
  </div>ð
  <div class="game-grid" id="gameGrid">ð
    <!-- Populated by JS -->ð
  </div>ð
</div>ð
ð
<!-- Game container -->ð
<div id="gamePage" class="game-page" style="display:none">ð
  <div class="game-header" id="gameHeader">ð
    <h2 id="gameTitle"></h2>ð
    <div class="game-controls" id="gameControls"></div>ð
  </div>ð
  <div class="game-area" id="gameArea">ð
    <!-- Each game renders here -->ð
  </div>ð
</div>ð
</main>ð
ð
<footer class="app-footer">ð
  <p>vÊfz\ n8b T -Æ - --eöges@ÿ </p>ð
</footer>ð
</div>ð
ð
<script src="js/games/memory.js"></script>ð
<script src="js/games/snake.js"></script>ð
<script src="js/games/2048.js"></script>ð
<script src="js/games/minesweeper.js"></script>ð
<script src="js/games/puzzle.js"></script>ð
<script src="js/games/tetris.js"></script>ð
<script src="js/games/tictactoe.js"></script>ð
<script src="js/games/numberguess.js"></script>ð
<script src="js/games/breakout.js"></script>ð
<script src="js/games/reaction.js"></script>ð
<script src="js/games/sudoku.js"></script>ð
<script src="js/games/flappy.js"></script>ð
<script src="js/games/fruitcatch.js"></script>ð
<script src="js/games/colormatch.js"></script>ð
<script src="js/games/delta.js"></script>ð
<script src="js/games/bubble.js"></script>ð
```


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```
<script src="js/games/invaders.js"></script>ð
<script src="js/games/wordle.js"></script>ð
<script src="js/games/simon.js"></script>ð
<script src="js/games/whack.js"></script>ð
<script src="js/games/match3.js"></script>ð
<script src="js/app.js"></script>ð
</body>ð
</html>ð
```

===== js/games/breakout.js =====

=====

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=====

```
/**
 * Breakout (bSx WW) - 5 Qs•İQsj!_
 * @author ch2077
 * @version 2.2.0
 * @date 2025-04-18 R Qe5Qs•İQsÿ kİQsx WWc'^ N T
 * @date 2025-05-03 Æ N†c!g•_9t P0Yû{-lõÿ hitPosf \ dxfô•êq6
 *
 * ~İQxbSx Wwn8b ÿ c!g•Sİ_9\ t m^-db@g x WWSs•ÇQs
 * Sİ_9%0^!u1t QÜN-c!g•v„OM•nQ³[šÿ -0.5~0.5f \ R0-4~4ÿ
 * kİQsx WW^Lep0 R ep0 t • • Xžÿ {,5Qs7^L10R +•...š0•
 * u T}3gay c%t bcT}ÿ T}\=game over
 */
const BreakoutGame = {
  canvas: null, ctx: null,
  width: 480, height: 320,
  paddle: { x: 200, w: 80, h: 12 }, // c!g•[½80š012
  ball: { x: 240, y: 280, r: 6, dx: 3, dy: -3 }, // t SJ_„6ÿ R YĖeæN
  bricks: [],
  score: 0,
  lives: 3,
  level: 1,
  maxLevel: 5,
  running: false,
  loop: null,
  container: null,
  statsCb: null,

  levelConfigs: [
    { rows: 4, cols: 7, speed: 3, colors: ['#ff6b6b','#ffa726','#ffd93d','#
00d2a0'] },
    { rows: 5, cols: 8, speed: 3.5, colors: ['#ff6b6b','#ffa726','#ffd93d',
-----
```

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```
'#00d2a0','#54a0ff'] },
  { rows: 5, cols: 9, speed: 4, colors: ['#ff6b6b','#ffa726','#ffd93d','#
00d2a0','#54a0ff'] },
  { rows: 6, cols: 9, speed: 4.5, colors: ['#ff6b6b','#ffa726','#ffd93d',
'#00d2a0','#54a0ff','#a29bfe'] },
  { rows: 7, cols: 10, speed: 5, colors: ['#ff6b6b','#ffa726','#ffd93d','#
#00d2a0','#54a0ff','#a29bfe','#ee5a24'] },
],
```

```
init(container, statsCb) {
  this.container = container;
  this.statsCb = statsCb;
  this.score = 0;
  this.lives = 3;
  this.level = 1;
  this.running = true;

  const w = Math.min(480, container.clientWidth - 48);
  const h = w * 0.67;
  this.width = w;
  this.height = h;

  this.canvas = document.createElement('canvas');
```

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```
this.canvas.className = 'game-canvas';
this.canvas.width = w;
this.canvas.height = h;
this.ctx = this.canvas.getContext('2d');

this.paddle = { x: w / 2 - 40, w: 80, h: 12 };

container.innerHTML = '';
container.appendChild(this.canvas);
const inst = document.createElement('div');
inst.className = 'instructions';
inst.textContent = '!• !' yûR`c!g• | Ÿ h / %æ\ObÖR`;
container.appendChild(inst);

this.showLevelIntro(() => {
  this.buildLevel();
  this.updateStats();
  document.addEventListener('keydown', this.handleKey);
  document.addEventListener('mousemove', this.handleMouse);
```

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```

    this.canvas.addEventListener('touchmove', this.handleTouch, {passive:
false});
    this.canvas.addEventListener('touchstart', this.handleTouch, {passive
:false});
    this.loop = setInterval(() => this.tick(), 16);
  });
},

buildLevel() {
  const cfg = this.levelConfigs[this.level - 1];
  const cols = cfg.cols, rows = cfg.rows;
  const bw = (this.width - 16) / cols, bh = 16;
  this.bricks = [];
  for (let r = 0; r < rows; r++)
    for (let c = 0; c < cols; c++)
      this.bricks.push({
        x: c * bw + 8, y: r * (bh + 4) + 30,
        w: bw - 4, h: bh, alive: true,
        color: cfg.colors[r % cfg.colors.length]
      });
  this.ball = {
    x: this.width / 2, y: this.height - 40, r: 6,
    dx: (Math.random() - 0.5) * cfg.speed, dy: -cfg.speed
  };
},

showLevelIntro(cb) {
  const div = document.createElement('div');
  div.className = 'level-intro';
  div.innerHTML = `<div class="level-num">{, ${this.level} Qs</div><div cla
ss="level-title">bSx WW</div>`;
  this.container.style.position = 'relative';
  this.container.appendChild(div);
  setTimeout(() => { if (div.parentNode) div.remove(); cb(); }, 1500);
}
```

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},

```

showLevelClear(cb) {
  this.running = false;
  clearInterval(this.loop);
  const div = document.createElement('div');
  div.className = 'level-intro';
  div.innerHTML = `<div class="level-num">' •ÇQsy </div><div class="level-ti
-----
```

```
=====
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tle">QÆY •ÛQe{, ${this.level + 1} Qs</div>`;
  this.container.appendChild(div);
  setTimeout(() => { if (div.parentNode) div.remove(); cb(); }, 1500);
},

handleKey(e) {
  if (e.key === 'ArrowLeft') BreakoutGame.paddle.x -= 20;
  if (e.key === 'ArrowRight') BreakoutGame.paddle.x += 20;
  BreakoutGame.paddle.x = Math.max(0, Math.min(BreakoutGame.width - Break
outGame.paddle.w, BreakoutGame.paddle.x));
},

handleMouse(e) {
  const rect = BreakoutGame.canvas.getBoundingClientRect();
  BreakoutGame.paddle.x = e.clientX - rect.left - BreakoutGame.paddle.w /
2;
  BreakoutGame.paddle.x = Math.max(0, Math.min(BreakoutGame.width - Break
outGame.paddle.w, BreakoutGame.paddle.x));
},

handleTouch(e) {
  e.preventDefault();
  const rect = BreakoutGame.canvas.getBoundingClientRect();
  BreakoutGame.paddle.x = e.touches[0].clientX - rect.left - BreakoutGame
.paddle.w / 2;
  BreakoutGame.paddle.x = Math.max(0, Math.min(BreakoutGame.width - Break
outGame.paddle.w, BreakoutGame.paddle.x));
},

tick() {
  if (!this.running) return;
  const b = this.ball, p = this.paddle, spd = this.levelConfigs[this.leve
l - 1].speed;

  b.x += b.dx;
  b.y += b.dy;

  if (b.x - b.r <= 0 || b.x + b.r >= this.width) b.dx *= -1;
  if (b.y - b.r <= 0) b.dy *= -1;

  if (b.y + b.r >= this.height - p.h && b.y + b.r <= this.height - p.h +
10 &&
    b.x > p.x && b.x < p.x + p.w) {
    b.dy = -Math.abs(b.dy);
    const hitPos = (b.x - p.x) / p.w;
    b.dx = (hitPos - 0.5) * spd * 2;
    if (Math.abs(b.dy) < spd) b.dy = -spd;
  }
}
```

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```
if (b.y - b.r > this.height) {
```

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{, 11 ~u

```
this.lives--;
this.updateStats();
if (this.lives <= 0) { this.gameOver(); return; }
b.x = this.width / 2; b.y = this.height - 40;
b.dx = (Math.random() - 0.5) * spd; b.dy = -spd;
}
```

```
for (const brick of this.bricks) {
  if (!brick.alive) continue;
  if (b.x + b.r > brick.x && b.x - b.r < brick.x + brick.w &&
    b.y + b.r > brick.y && b.y - b.r < brick.y + brick.h) {
    brick.alive = false;
    b.dy *= -1;
    this.score += 10;
    this.updateStats();
    if (this.bricks.every(br => !br.alive)) {
      if (this.level >= this.maxLevel) { this.gameOver(true); }
      else { this.levelClear(); }
      return;
    }
    break;
  }
}
```

```
this.draw();
},
```

```
levelClear() {
  this.running = false;
  clearInterval(this.loop);
  document.removeEventListener('keydown', this.handleKey);
  document.removeEventListener('mousemove', this.handleMouse);
  const div = document.createElement('div');
  div.className = 'level-intro';
  div.innerHTML = `<div class="level-num">'  {, ${this.level} Qs• •Çÿ </div><di
v class="level-title">QÆY •ÛQe{, ${this.level + 1} Qs</div>`;
  this.container.appendChild(div);
  setTimeout(() => {
    if (div.parentNode) div.remove();
    this.level++;
  }, 1000);
}
```

=====

vÊfz\ n8b T -Æ^sSð|û-ß V3.2

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{, 14 ~u

```
    this.running = true;
    document.addEventListener('keydown', this.handleKey);
    document.addEventListener('mousemove', this.handleMouse);
    this.canvas.addEventListener('touchmove', this.handleTouch, {passive:
false});
    this.canvas.addEventListener('touchstart', this.handleTouch, {passive
:false});
    this.showLevelIntro(() => {
        this.buildLevel();
        this.updateStats();
        this.loop = setInterval(() => this.tick(), 16);
    });
}, 1800);
```

vÊfz\ n8b T -Æ^sSð|û-ß V3.2

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{, 12 ~u

```
    },
    draw() {
        const c = this.ctx;
        c.clearRect(0, 0, this.width, this.height);
        for (const b of this.bricks) {
            if (!b.alive) continue;
            c.fillStyle = b.color;
            c.fillRect(b.x, b.y, b.w, b.h);
        }
        c.fillStyle = '#a29bfe';
        c.fillRect(this.paddle.x, this.height - this.paddle.h, this.paddle.w, t
his.paddle.h);
        c.fillStyle = 'ffff';
        c.beginPath();
        c.arc(this.ball.x, this.ball.y, this.ball.r, 0, Math.PI * 2);
        c.fill();
    },
    updateStats() {
        if (this.statsCb) {
            this.statsCb({ 'QsSa': `${this.level}/${this.maxLevel}`, '_R ': this.sco
re, 'u T': this.lives });
        }
    },
    gameOver(won = false) {
        this.running = false;
        clearInterval(this.loop);
    }
}
```

vÊfz\ n8b T -Æ^sSð|û-ß V3.2

{, 15 ~u

```
document.removeEventListener('keydown', this.handleKey);
document.removeEventListener('mousemove', this.handleMouse);
const div = document.createElement('div');
div.className = 'game-over-overlay';
div.innerHTML = won
? `<h3>ð<ß%`mUæ• Qsŷ </h3><p>• •ÇN†Qh•è ${this.maxLevel} Qsŷ _-R : ${this.score}</p>
<button class="btn" id="replayBtn">Q•geN \@</button>`
: `<h3>n8b -ôg_</h3><p>{, ${this.level}/${this.maxLevel} Qs | _-R : ${this.sc
ore}</p><button class="btn" id="replayBtn">Q•geN \@</button>`;
this.container.style.position = 'relative';
this.container.appendChild(div);
document.getElementById('replayBtn').addEventListener('click', () => {
  this.destroy();
  this.init(this.container, this.statsCb);
});
},

destroy() {
  this.running = false;
  clearInterval(this.loop);
  document.removeEventListener('keydown', this.handleKey);
  document.removeEventListener('mousemove', this.handleMouse);
}
};
```

vÊfz\ n8b T -Æ^sSð|û-ß V3.2

{, 13 ~u

===== js/games/bubble.js =====

```
/**
 * Bubble Shooter (lálá\ bK) - 5 Qs•iQsj!_
 * @author ch2077
 * @version 3.1.0
 * @date 2025-04-28 [EQh'ÍQ™S9'M•;•'ŷ e9N:•Qh<•»c¥
 * @date 2025-05-08 OîY spawnBubble~æ,rN W(RiOYlálÁN-v„bug
 *
 * ~İQxláláŸ™s@lôŷ hex grid^ \@ŷ Pvep^L•)•ÛSJN*h<[Pŷ
 * S9'MhÂmKŷ flood fillc Qm%Ô•»c¥b-T ,r•p• R 'İŷ "e3m^-d
 * mnR`láláhÂmKŷ BFSNÎ~v^L•MS†ŷ N •p• v„c%„=^vR R
 * Qs•.OîY ŷ spawnBubbles°W(SêNÎRiOYlálá~æ,rl`N--•g:
 */
const BubbleGame = {
  canvas: null, ctx: null,
  bubbles: [], shooter: { x: 0, y: 0 },
```

=====

vÊfz\ n8b T -Æ^sSð|û~ß V3.2

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{, 16 ~u

```
currentBubble: null,
// Qm,rläläy c QsSa%ã• ~æ,repÿ 4!'4!'5!'5!'6ÿ
allColors: ['#ff6b6b','#ffa726','#ffd93d','#00d2a0','#54a0ff','#a29bfe'],
colors: [],
score: 0, shots: 0,
rows: 8, cols: 10,
cellSize: 0, radius: 0,
container: null, statsCb: null,
animFrame: null, running: false,
aimLine: { angle: Math.PI / 2 },
nextId: 0,
level: 1, maxLevel: 5,
levelConfigs: [
  { rows: 5, numColors: 4 },
  { rows: 6, numColors: 4 },
  { rows: 7, numColors: 5 },
  { rows: 8, numColors: 5 },
  { rows: 9, numColors: 6 },
],
init(container, statsCb) {
  this.container = container;
  this.statsCb = statsCb;
  this.score = 0;
  this.shots = 0;
  this.level = 1;
  this.running = true;
  this.currentBubble = null;
  this.bubbles = [];
  this.nextId = 0;
  this.render();
  this.setupCanvas();
  this.showLevelIntro(() => {
```

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{, 14 ~u

```
    this.startLevel();
    this.loop();
  });
},
startLevel() {
  const cfg = this.levelConfigs[this.level - 1];
  this.colors = this.allColors.slice(0, cfg.numColors);
  this.initGrid(cfg.rows);
}
```

```

    this.spawnBubble();
    this.updateStats();
  },

  render() {
    this.container.innerHTML = `<canvas id="bubbleCanvas" style="width:100%
;max-width:400px;border-radius:8px;touch-action:none"></canvas>`;
  },

  setupCanvas() {
    this.canvas = document.getElementById('bubbleCanvas');
    this.ctx = this.canvas.getContext('2d');
    const w = Math.min(400, this.container.clientWidth - 32);
    this.canvas.width = w;
    this.canvas.height = w * 1.5;
    this.cellSize = this.canvas.width / this.cols;
    this.radius = this.cellSize * 0.42;
    this.shooter = { x: this.canvas.width / 2, y: this.canvas.height - this
.radius - 4 };

    const updateAim = (clientX, clientY) => {
      const rect = this.canvas.getBoundingClientRect();
      const mx = (clientX - rect.left) * (this.canvas.width / rect.width);
      const my = (clientY - rect.top) * (this.canvas.height / rect.height);
      const dx = mx - this.shooter.x;
      const dy = my - this.shooter.y;
      if (dy < 0) this.aimLine.angle = Math.atan2(-dy, dx);
    };

    this.canvas.addEventListener('mousemove', e => { updateAim(e.clientX, e
.clientY); });
    this.canvas.addEventListener('touchmove', e => { e.preventDefault(); up
dateAim(e.touches[0].clientX, e.touches[0].clientY); });
    this.canvas.addEventListener('click', () => { this.shoot(); });
    this.canvas.addEventListener('touchend', () => { this.shoot(); });
  },

  initGrid(numRows) {
    this.bubbles = [];
    for (let r = 0; r < numRows; r++) {
      const offset = (r % 2) * this.cellSize / 2;
      const maxCols = (r % 2 === 1) ? this.cols - 1 : this.cols;
      for (let c = 0; c < maxCols; c++) {
        this.bubbles.push({
          id: this.nextId++,

```

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vÊfz\ n8b T -Æ^sSð|û~ß V3.2

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{, 18 ~u

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{, 15 ~u

=====

```
    x: offset + c * this.cellSize + this.cellSize / 2,
    y: r * this.cellSize * 0.87 + this.cellSize / 2 + 10,
    r: this.radius,
    color: this.colors[Math.floor(Math.random() * this.colors.length)
],
    row: r, col: c
  });
}
},
showLevelIntro(cb) {
  const div = document.createElement('div');
  div.className = 'level-intro';
  div.innerHTML = `<div class="level-num">{, ${this.level} Qs</div><div cla
ss="level-title">lálá\ bK · ${this.levelConfigs[this.level-1].numColors} ,r</div
>`;
  this.container.style.position = 'relative';
  this.container.appendChild(div);
  setTimeout(() => { if (div.parentNode) div.remove(); cb(); }, 1500);
},
spawnBubble() {
  // Only pick colors that still exist on the grid (or any if grid is emp
ty)
  let pool = this.colors;
  if (this.bubbles.length > 0) {
    const remainingColors = [...new Set(this.bubbles.map(b => b.color))];
    if (remainingColors.length > 0) pool = remainingColors;
  }
  this.currentBubble = {
    x: this.shooter.x, y: this.shooter.y,
    r: this.radius,
    color: pool[Math.floor(Math.random() * pool.length)],
    vx: 0, vy: 0, active: true, shooting: false
  };
},
shoot() {
  if (!this.currentBubble || this.currentBubble.shooting) return;
  const speed = 14;
  this.currentBubble.vx = Math.cos(this.aimLine.angle) * speed;
  this.currentBubble.vy = -Math.sin(this.aimLine.angle) * speed;
  this.currentBubble.shooting = true;
  this.shots++;
}
```

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{, 19 ~u

```
},
update() {
  const cb = this.currentBubble;
  if (!cb || !cb.shooting) return;

  cb.x += cb.vx;
  cb.y += cb.vy;
```

=====

vÊfz\ n8b T -Æ^sSð|û~ß V3.2

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{, 16 ~u

```
if (cb.x - cb.r <= 0 || cb.x + cb.r >= this.canvas.width) {
  cb.vx *= -1;
  cb.x = Math.max(cb.r, Math.min(this.canvas.width - cb.r, cb.x));
}

if (cb.y - cb.r <= 0) {
  this.snapBubble(cb);
  return;
}

for (let i = 0; i < this.bubbles.length; i++) {
  const b = this.bubbles[i];
  const dx = cb.x - b.x;
  const dy = cb.y - b.y;
  if (Math.sqrt(dx * dx + dy * dy) < this.radius * 2 * 0.95) {
    this.snapBubble(cb);
    return;
  }
}

if (cb.y > this.canvas.height + this.radius) {
  cb.shooting = false; cb.active = false;
  this.spawnBubble();
}
},

snapBubble(bubble) {
  bubble.active = false; bubble.shooting = false; bubble.id = this.nextId
  ++;

  let bestR = 0, bestC = 0, bestDist = Infinity;
  for (let r = 0; r < this.rows; r++) {
    const offset = (r % 2) * this.cellSize / 2;
```

=====

vÊfz\ n8b T -Æ^sSð|û~ß V3.2

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{, 20 ~u

```
const maxCols = (r % 2 === 1) ? this.cols - 1 : this.cols;
for (let c = 0; c < maxCols; c++) {
  const gx = offset + c * this.cellSize + this.cellSize / 2;
  const gy = r * this.cellSize * 0.87 + this.cellSize / 2 + 10;
  const dx = bubble.x - gx, dy = bubble.y - gy;
  const d = dx * dx + dy * dy;
  if (d < bestDist) {
    if (!this.bubbles.some(b => b.row === r && b.col === c)) {
      bestDist = d; bestR = r; bestC = c;
    }
  }
}

bubble.row = bestR; bubble.col = bestC;
const offset = (bestR % 2) * this.cellSize / 2;
bubble.x = offset + bestC * this.cellSize + this.cellSize / 2;
bubble.y = bestR * this.cellSize * 0.87 + this.cellSize / 2 + 10;
```

vÊfz\ n8b T -Æ^sSð|û~ß V3.2

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{, 17 ~u

```
this.bubbles.push(bubble);

const matched = this.findMatches(bubble);
if (matched.length >= 3) {
  this.score += matched.length * 10;
  const matchedIds = new Set(matched.map(b => b.id));
  this.bubbles = this.bubbles.filter(b => !matchedIds.has(b.id));
  this.removeFloating();
}

this.spawnBubble();
this.updateStats();

// Level clear or game over
if (this.bubbles.length === 0) {
  if (this.level >= this.maxLevel) { this.gameOver(true); }
  else { this.levelClear(); }
  return;
}
const loseRow = this.bubbles.some(b => b.y + b.r > this.shooter.y - 10)
;
  if (loseRow) this.gameOver(false);
},
```

```
getNeighbors(bubble) {
  const r = bubble.row, c = bubble.col;
  const candidates = [];
  if (r % 2 === 0) {
    candidates.push({ row: r, col: c - 1 }, { row: r, col: c + 1 },
      { row: r - 1, col: c - 1 }, { row: r - 1, col: c },
      { row: r + 1, col: c - 1 }, { row: r + 1, col: c });
  } else {
    candidates.push({ row: r, col: c - 1 }, { row: r, col: c + 1 },
      { row: r - 1, col: c }, { row: r - 1, col: c + 1 },
      { row: r + 1, col: c }, { row: r + 1, col: c + 1 });
  }
  const neighbors = [];
  for (const pos of candidates) {
    const b = this.bubbles.find(b => b.row === pos.row && b.col === pos.c
ol);
    if (b && b.color === bubble.color) neighbors.push(b);
  }
  return neighbors;
},

findMatches(bubble) {
  const visited = new Set(), matches = [], stack = [bubble];
  while (stack.length) {
    const b = stack.pop();
    if (visited.has(b.id)) continue;
    visited.add(b.id); matches.push(b);
    for (const nb of this.getNeighbors(b)) {
```

```
    if (!visited.has(nb.id)) stack.push(nb);
  }
}
return matches;
},

removeFloating() {
  const connected = new Set();
  const queue = this.bubbles.filter(b => b.row <= 0);
  queue.forEach(b => connected.add(b.id));
  while (queue.length) {
    const b = queue.shift();
    const r = b.row, c = b.col;
    const candidates = [];
```

```
if (r % 2 === 0) {
  candidates.push({ row: r, col: c - 1 }, { row: r, col: c + 1 },
    { row: r - 1, col: c - 1 }, { row: r - 1, col: c },
    { row: r + 1, col: c - 1 }, { row: r + 1, col: c });
} else {
  candidates.push({ row: r, col: c - 1 }, { row: r, col: c + 1 },
    { row: r - 1, col: c }, { row: r - 1, col: c + 1 },
    { row: r + 1, col: c }, { row: r + 1, col: c + 1 });
}
for (const pos of candidates) {
  const nb = this.bubbles.find(b => b.row === pos.row && b.col === po
s.col);
  if (nb && !connected.has(nb.id)) { connected.add(nb.id); queue.push
(nb); }
}
const before = this.bubbles.length;
this.bubbles = this.bubbles.filter(b => connected.has(b.id));
this.score += (before - this.bubbles.length) * 5;
},

levelClear() {
  this.running = false;
  cancelAnimationFrame(this.animFrame);
  const div = document.createElement('div');
  div.className = 'level-intro';
  div.innerHTML = `<div class="level-num"> {, ${this.level} Qs• •Çÿ </div><di
v class="level-title">QÆY •ÛQe{, ${this.level + 1} Qs</div>`;
  this.container.appendChild(div);
  setTimeout(() => {
    if (div.parentNode) div.remove();
    this.level++;
    this.running = true;
    this.bubbles = [];
    this.currentBubble = null;
    this.showLevelIntro(() => {
      this.startLevel();
      this.loop();
    });
  });
}
```

```
}, 1800);
},
```

```
draw() {
  const ctx = this.ctx;
  ctx.clearRect(0, 0, this.canvas.width, this.canvas.height);
  for (const b of this.bubbles) this.drawBubble(b.x, b.y, b.r, b.color);
  if (this.currentBubble && !this.currentBubble.shooting) {
    ctx.strokeStyle = 'rgba(255,255,255,0.3)';
    ctx.lineWidth = 1; ctx.setLineDash([4, 4]);
    ctx.beginPath();
    ctx.moveTo(this.shooter.x, this.shooter.y);
    ctx.lineTo(this.shooter.x + Math.cos(this.aimLine.angle) * 80, this.s
hooter.y - Math.sin(this.aimLine.angle) * 80);
    ctx.stroke(); ctx.setLineDash([]);
  }
  if (this.currentBubble) this.drawBubble(this.currentBubble.x, this.curr
entBubble.y, this.currentBubble.r, this.currentBubble.color);
},

drawBubble(x, y, r, color) {
  const ctx = this.ctx;
  ctx.beginPath(); ctx.arc(x, y, r, 0, Math.PI * 2);
  ctx.fillStyle = color; ctx.fill();
  ctx.strokeStyle = 'rgba(255,255,255,0.25)'; ctx.lineWidth = 1; ctx.stro
ke();
  ctx.beginPath(); ctx.arc(x - r * 0.3, y - r * 0.3, r * 0.3, 0, Math.PI
* 2);
  ctx.fillStyle = 'rgba(255,255,255,0.3)'; ctx.fill();
},

loop() {
  if (!this.running) return;
  this.update(); this.draw();
  this.animFrame = requestAnimationFrame(() => this.loop());
},

updateStats() {
  if (this.statsCb) this.statsCb({ 'QsSa': `${this.level}/${this.maxLevel}`
, '_R ': this.score });
},

gameOver(won = false) {
  this.running = false;
  cancelAnimationFrame(this.animFrame);
  const div = document.createElement('div');
  div.className = 'game-over-overlay';
  div.innerHTML = won
    ? `<h3>ð<ß% `mUæ• Qsŷ </h3><p>• •ÇN†Qh•è ${this.maxLevel} Qsŷ _R : ${this.score}</p>`
    : `<h3>n8b ~ôg_</h3><p>{, ${this.level}/${this.maxLevel} Qs | _R : ${this.sc
```

vÊfz\ n8b T -Æ^sSð|û-ß V3.2 {, 24 ~u

```
ore}</p><button class="btn" id="replayBtn">Q•geN k!</button>`;
  this.container.style.position = 'relative';
  this.container.appendChild(div);
  document.getElementById('replayBtn').addEventListener('click', () => {
    this.destroy(); this.init(this.container, this.statsCb);
  });
```

vÊfz\ n8b T -Æ^sSð|û-ß V3.2 {, 20 ~u

```
},
destroy() {
  this.running = false;
  cancelAnimationFrame(this.animFrame);
}
};
```

===== js/games/invaders.js =====

```
/**
 * Space Invaders (Y*zz0µue€ ) - 5 lâ•iQsj!_
 * @author ch2077
 * @version 2.0.1
 * @date 2025-04-12 'ÍQ™N:5lâ•iQsÿ kİlân T ,0- 'M•n
 * @date 2025-04-25 R N†•ÇQsYVR±u T}g:R6ÿ • QsVpná
 *
 * ~İQx^wg:\ Qûÿ m^pmb@g Y f ,0- •ÛQeN N lâ
 * Y f N°• - ]æSóyûR`%æ•¹N yûÿ -•g:T s@[¶\ Qû
 * 5lâ-¼^!• Xžÿ O! [ß!'z•Qû!' |¼,ñ!'s<rL!'BOSS
 * e/c •.v0e¹T •.+ÿ h yûR`+%æ\ObÖR`cSR6`p,9
 */
```

```
const InvadersGame = {
  canvas: null, ctx: null,
  player: { x: 0, y: 0, w: 40, h: 30 },
  aliens: [], bullets: [], alienBullets: [],
  alienDir: 1, alienSpeed: 0.5, alienDrop: 8,
  score: 0, lives: 3, level: 1, maxLevel: 5,
  running: false, animFrame: null,
  container: null, statsCb: null,
  moveLeft: false, moveRight: false,
  lastShot: 0, shotCooldown: 400,
  transitioning: false,

  levelConfigs: [
    { rows: 3, speed: 0.5, shootRate: 0.01, name: 'O! [ß,0- ' },
```


=====

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{, 25 ~u

```
{ rows: 4, speed: 0.7, shootRate: 0.015, name: 'z•Qû,0- ' },
{ rows: 5, speed: 0.9, shootRate: 0.02, name: '|%,ñ,0- ' },
{ rows: 5, speed: 1.1, shootRate: 0.025, name: 's<rL,0- ' },
{ rows: 6, speed: 1.3, shootRate: 0.03, name: 'BOSS,0- ' },
],
```

```
init(container, statsCb) {
  this.container = container;
  this.statsCb = statsCb;
  this.score = 0; this.lives = 3; this.level = 1;
  this.bullets = []; this.alienBullets = [];
  this.alienDir = 1;
  this.moveLeft = false; this.moveRight = false;
  this.running = true; this.transitioning = false;
```

=====

vĖfz\ n8b T -Æ^sSð|û-ß V3.2

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{, 21 ~u

```
this.render();
this.setupCanvas();
this.bindControls();
this.showLevelIntro(() => {
  this.spawnAliens();
  this.updateStats();
  this.loop();
});
},
```

```
render() {
  this.container.innerHTML = `<canvas id="invCanvas" style="width:100%;ma
x-width:400px;border-radius:8px;touch-action:none"></canvas>`;
},
```

```
setupCanvas() {
  this.canvas = document.getElementById('invCanvas');
  this.ctx = this.canvas.getContext('2d');
  const w = Math.min(400, this.container.clientWidth - 32);
  this.canvas.width = w;
  this.canvas.height = w * 1.4;
  this.player.x = this.canvas.width / 2;
  this.player.y = this.canvas.height - 50;
},
```

```
showLevelIntro(cb) {
  this.transitioning = true;
  const cfg = this.levelConfigs[this.level - 1];
```

=====

vĖfz\ n8b T -Æ^sSð|û-ß V3.2

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{, 26 ~u

```
const div = document.createElement('div');
div.className = 'level-intro';
div.innerHTML = `<div class="level-num">{, ${this.level} Qs</div><div cla
ss="level-title">${cfg.name}</div>`;
this.container.style.position = 'relative';
this.container.appendChild(div);
setTimeout(() => {
  if (div.parentNode) div.remove();
  this.transitioning = false;
  cb();
}, 1500);
},
```

```
spawnAliens() {
  const cfg = this.levelConfigs[this.level - 1];
  this.alienSpeed = cfg.speed;
  this.alienDir = 1;
  this.aliens = [];
  const cols = 8, rows = cfg.rows;
  const w = this.canvas.width / (cols + 1);
  for (let r = 0; r < rows; r++) {
    for (let c = 0; c < cols; c++) {
      this.aliens.push({
        x: w + c * w, y: 40 + r * 40,
```

vĖfz\ n8b T -Æ^sSð|û-ß V3.2

=====

{, 22 ~u

```
      w: 24, h: 20, alive: true,
      type: r % 3
    });
  }
}
},

bindControls() {
  const keydown = (e) => {
    if (e.key === 'ArrowLeft' || e.key === 'a') this.moveLeft = true;
    if (e.key === 'ArrowRight' || e.key === 'd') this.moveRight = true;
  };
  const keyup = (e) => {
    if (e.key === 'ArrowLeft' || e.key === 'a') this.moveLeft = false;
    if (e.key === 'ArrowRight' || e.key === 'd') this.moveRight = false;
  };
  document.addEventListener('keydown', keydown);
  document.addEventListener('keyup', keyup);
}
```

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vÊfz\ n8b T -Æ^sSð|û~ß V3.2

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{, 27 ~u

```
this._keydown = keydown; this._keyup = keyup;

this.canvas.addEventListener('touchmove', e => {
  e.preventDefault();
  const rect = this.canvas.getBoundingClientRect();
  const tx = (e.touches[0].clientX - rect.left) * (this.canvas.width /
rect.width);
  this.player.x = Math.max(this.player.w/2, Math.min(this.canvas.width
- this.player.w/2, tx));
});
this.canvas.addEventListener('touchstart', e => {
  e.preventDefault();
  const rect = this.canvas.getBoundingClientRect();
  const tx = (e.touches[0].clientX - rect.left) * (this.canvas.width /
rect.width);
  this.player.x = Math.max(this.player.w/2, Math.min(this.canvas.width
- this.player.w/2, tx));
  this.shoot();
});
this.canvas.addEventListener('click', () => this.shoot());
this.canvas.addEventListener('mousemove', e => {
  const rect = this.canvas.getBoundingClientRect();
  const mx = (e.clientX - rect.left) * (this.canvas.width / rect.width)
;
  this.player.x = Math.max(this.player.w/2, Math.min(this.canvas.width
- this.player.w/2, mx));
});
},

shoot() {
  const now = Date.now();
  if (now - this.lastShot < this.shotCooldown) return;
  this.lastShot = now;
  this.bullets.push({ x: this.player.x, y: this.player.y - 20, vy: -6 });
},

update() {
  if (this.transitioning) return;
```

=====

vÊfz\ n8b T -Æ^sSð|û~ß V3.2

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{, 23 ~u

```
if (this.moveLeft) this.player.x -= 5;
if (this.moveRight) this.player.x += 5;
this.player.x = Math.max(this.player.w/2, Math.min(this.canvas.width -
this.player.w/2, this.player.x));

-----
```

```
const now = Date.now();
if (now - this.lastShot > 800) this.shoot();

for (const b of this.bullets) b.y += b.vy;
this.bullets = this.bullets.filter(b => b.y > -10);

for (const b of this.alienBullets) b.y += 2.5;
this.alienBullets = this.alienBullets.filter(b => b.y < this.canvas.height + 10);

const cfg = this.levelConfigs[this.level - 1];
const liveAliens = this.aliens.filter(a => a.alive);
if (liveAliens.length && Math.random() < cfg.shootRate) {
  const shooter = liveAliens[Math.floor(Math.random() * liveAliens.length)];
  this.alienBullets.push({ x: shooter.x, y: shooter.y + 10 });
}

let hitEdge = false;
for (const a of this.aliens) {
  if (!a.alive) continue;
  a.x += this.alienDir * this.alienSpeed;
  if (a.x + a.w/2 >= this.canvas.width || a.x - a.w/2 <= 0) hitEdge = true;
}
if (hitEdge) {
  this.alienDir *= -1;
  for (const a of this.aliens) {
    if (!a.alive) continue;
    a.y += this.alienDrop;
    a.x += this.alienDir * this.alienSpeed;
  }
}

for (const b of this.bullets) {
  for (const a of this.aliens) {
    if (!a.alive) continue;
    if (Math.abs(b.x - a.x) < a.w/2 && Math.abs(b.y - a.y) < a.h/2) {
      a.alive = false; b.y = -999;
      this.score += (a.type + 1) * 10;
    }
  }
}
this.bullets = this.bullets.filter(b => b.y > -10 && b.y < this.canvas.height + 10);

for (const b of this.alienBullets) {
```

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vÊfz\ n8b T -Æ^sSð|û~ß V3.2

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{, 29 ~u

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```
    if (Math.abs(b.x - this.player.x) < this.player.w/2 && Math.abs(b.y -
this.player.y) < this.player.h/2) {
        b.y = -999; this.lives--;
        if (this.lives <= 0) { this.gameOver(); return; }
```

vÊfz\ n8b T -Æ^sSð|û~ß V3.2

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{, 24 ~u

=====

```
    }
}
this.alienBullets = this.alienBullets.filter(b => b.y > -10);

if (this.aliens.some(a => a.alive && a.y + a.h/2 >= this.player.y)) {
    this.lives = 0; this.gameOver(); return;
}

if (this.aliens.every(a => !a.alive)) {
    if (this.level >= this.maxLevel) { this.gameOver(true); return; }
    this.level++;
    this.lives = Math.min(5, this.lives + 1); // Bonus life
    this.updateStats();
    this.showLevelIntro(() => {
        this.spawnAliens();
    });
}

this.updateStats();
},

draw() {
    const ctx = this.ctx;
    ctx.clearRect(0, 0, this.canvas.width, this.canvas.height);
    for (const a of this.aliens) {
        if (!a.alive) continue;
        const colors = ['#54a0ff', '#ffa726', '#ff6b6b'];
        ctx.fillStyle = colors[a.type];
        ctx.fillRect(a.x - a.w/2, a.y - a.h/2, a.w, a.h);
        ctx.fillStyle = 'fff';
        ctx.fillRect(a.x - 5, a.y - 5, 3, 4);
        ctx.fillRect(a.x + 2, a.y - 5, 3, 4);
    }
    ctx.fillStyle = '#00d2a0';
    ctx.beginPath();
    ctx.moveTo(this.player.x, this.player.y - this.player.h/2);
    ctx.lineTo(this.player.x - this.player.w/2, this.player.y + this.player
.h/2);
}

-----
```

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vÊfz\ n8b T -Æ^sSð|û-ß V3.2

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{, 30 ~u

```
    ctx.lineTo(this.player.x + this.player.w/2, this.player.y + this.player
.h/2);
    ctx.closePath(); ctx.fill();
    ctx.fillStyle = '#ffd93d';
    for (const b of this.bullets) ctx.fillRect(b.x - 2, b.y - 6, 4, 10);
    ctx.fillStyle = '#ff6b6b';
    for (const b of this.alienBullets) ctx.fillRect(b.x - 2, b.y - 4, 4, 8)
;
    ctx.fillStyle = 'rgba(255,255,255,0.5)';
    const starSeed = 42;
    for (let i = 0; i < 40; i++) {
        const sx = ((i * 137 + starSeed) % this.canvas.width);
        const sy = ((i * 251 + this.score * 10) % this.canvas.height);
        ctx.fillRect(sx, sy, 1.5, 1.5);
    }
}
```

vÊfz\ n8b T -Æ^sSð|û-ß V3.2

{, 25 ~u

```
    },
    loop() {
        if (!this.running) return;
        this.update(); this.draw();
        this.animFrame = requestAnimationFrame(() => this.loop());
    },
    updateStats() {
        if (this.statsCb) this.statsCb({ 'QsSa': `${this.level}/${this.maxLevel}`
, '_R ': this.score, 'u T': this.lives });
    },
    gameOver(won = false) {
        this.running = false;
        cancelAnimationFrame(this.animFrame);
        document.removeEventListener('keydown', this._keydown);
        document.removeEventListener('keyup', this._keyup);
        const div = document.createElement('div');
        div.className = 'game-over-overlay';
        div.innerHTML = won
? `<h3>ð<ß% `mUæ• Qsý </h3><p>m^pmN†Qh•è ${this.maxLevel} lâ,0- ý _R : ${this.score}</
p><button class="btn" id="replayBtn">Q•geN \@</button>`
: `<h3>n8b -ôg_</h3><p>R0•%{, ${this.level}/${this.maxLevel} Qs | _R : ${this.
score}</p><button class="btn" id="replayBtn">Q•geN k!</button>`;
        this.container.style.position = 'relative';
        this.container.appendChild(div);
    }
}
```

vĖfz\ n8b T -Æ^sSð|û~ß V3.2

{, 31 ~u

```
document.getElementById('replayBtn').addEventListener('click', () => {
  this.destroy(); this.init(this.container, this.statsCb);
});
},

destroy() {
  this.running = false;
  cancelAnimationFrame(this.animFrame);
  document.removeEventListener('keydown', this._keydown);
  document.removeEventListener('keyup', this._keyup);
}
};
```

===== js/games/snake.js =====

```
/**
 * Snake Game (*T †Ç) - 5 Qs•iQsj!_
 * @author ch2077
 * @version 2.1.0
 * @date 2025-03-15 'İg„yûR”zİ• 'M
 * @date 2025-04-20 XžR •iQs|û~ß
 *
 * ~İQx•*T †Ç[žs°ÿ 'Çu(tile-based•Qh<yûR”
 * yûR”zİPZN†swipebKRž + D-PadSİ• 'Mÿ -2kbe†T <İR$R N†15pxk{S:
 * • ^|-•QsSa• Xžÿ R YE100ms/keÿ kİQs-MON~|15-20ms
 */
```

vĖfz\ n8b T -Æ^sSð|û~ß V3.2

{, 26 ~u

```
const SnakeGame = {
  canvas: null, ctx: null,
  tileCount: 20, tileSize: 0, // 20x20h<ÿ W0Vp\:[ø•ê• ^”
  snake: [], food: null,
  dir: { x: 1, y: 0 }, nextDir: { x: 1, y: 0 },
  // læa ŷ nextDiru(Nž• Q²•“Qeÿ -2kbT N ^'Q…N$#!SÍT [ü•ô•êdž
  score: 0, speed: 100,
  loop: null, running: false,
  container: null, statsCb: null,
  touchStart: null,
  level: 1, maxLevel: 5,
  targetScore: 50,
  levelConfigs: [
    { target: 50, speed: 100, name: 'e°bKQe•è' },
    { target: 100, speed: 85, name: 'n Qe0sXf' },
    { target: 150, speed: 70, name: '• ^|c b ' },
  ],
};
```

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vĒfz\ n8b T -Æ^sSđ|û~ß V3.2

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{, 32 ~u

```
{ target: 200, speed: 55, name: 'g•-PSí^"' },
{ target: 300, speed: 45, name: '†Çs<NK•i' },
],

init(container, statsCb) {
  this.container = container;
  this.statsCb = statsCb;
  this.snake = [{ x: 10, y: 10 }];
  this.dir = { x: 1, y: 0 }; this.nextDir = { x: 1, y: 0 };
  this.score = 0; this.level = 1;
  const cfg = this.levelConfigs[0];
  this.speed = cfg.speed; this.targetScore = cfg.target;
  this.running = true; this.touchStart = null;

  const size = Math.min(400, this.container.clientWidth - 48);
  this.canvas = document.createElement('canvas');
  this.canvas.className = 'game-canvas';
  this.canvas.width = size; this.canvas.height = size;
  this.ctx = this.canvas.getContext('2d');
  this.tileSize = size / this.tileCount;

  this.container.innerHTML = '';
  this.container.appendChild(this.canvas);
  this.container.appendChild(this.buildDPad());
  this.spawnFood();
  this.showLevelIntro(() => {
    this.updateStats();
    document.addEventListener('keydown', this._handleKey);
    this.canvas.addEventListener('touchstart', this._touchStart, {passive
:false});
    this.canvas.addEventListener('touchend', this._touchEnd, {passive:fal
se});
    this.loop = setInterval(() => this.tick(), this.speed);
  });
},
```

=====

vĒfz\ n8b T -Æ^sSđ|û~ß V3.2

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{, 27 ~u

```
showLevelIntro(cb) {
  this.running = false;
  const cfg = this.levelConfigs[this.level - 1];
  const div = document.createElement('div');
  div.className = 'level-intro';
  div.innerHTML = `<div class="level-num">{, ${this.level} Qs</div><div cla
```

```
ss="level-title">${cfg.name} . vîh ${cfg.target} R </div>`;
  this.container.style.position = 'relative';
  this.container.appendChild(div);
  setTimeout(() => {
    if (div.parentNode) div.remove();
    this.running = true;
    cb();
  }, 1500);
},

_handleKey(e) {
  const k = e.key.toLowerCase();
  if (k === 'arrowup' || k === 'w') SnakeGame.setDir(0, -1);
  else if (k === 'arrowdown' || k === 's') SnakeGame.setDir(0, 1);
  else if (k === 'arrowleft' || k === 'a') SnakeGame.setDir(-1, 0);
  else if (k === 'arrowright' || k === 'd') SnakeGame.setDir(1, 0);
},

setDir(dx, dy) {
  if (this.dir.x === -dx && this.dir.y === -dy) return;
  this.nextDir = { x: dx, y: dy };
},

_touchStart(e) { e.preventDefault(); if(e.touches[0]) SnakeGame.touchStar
t={x:e.touches[0].clientX,y:e.touches[0].clientY}; },
_touchEnd(e) {
  e.preventDefault();
  if (!SnakeGame.touchStart) return;
  const dx = e.changedTouches[0].clientX - SnakeGame.touchStart.x;
  const dy = e.changedTouches[0].clientY - SnakeGame.touchStart.y;
  SnakeGame.touchStart = null;
  if (Math.abs(dx) < 15 && Math.abs(dy) < 15) return;
  if (Math.abs(dx) > Math.abs(dy)) SnakeGame.setDir(dx > 0 ? 1 : -1, 0);
  else SnakeGame.setDir(0, dy > 0 ? 1 : -1);
},

buildDPad() {
  const d = document.createElement('div');
  d.className = 'dpad';
  d.innerHTML = `
    <button class="dpad-btn dpad-up" id="dpUp">%²</button>
    <button class="dpad-btn dpad-left" id="dpLeft">%À</button>
    <button class="dpad-btn dpad-right" id="dpRight">%¶</button>
    <button class="dpad-btn dpad-down" id="dpDown">%¼</button>`;
  setTimeout(() => {
    const bind = (id, dx, dy) => {
```


=====

vÊfz\ n8b T -Æ^sSð|û-ß V3.2

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{, 35 ~u

```
const div = document.createElement('div');
div.className = 'level-intro';
div.innerHTML = `<div class="level-num">`  {, ${this.level} Qs• •Çÿ </div><di
v class="level-title">QÆY •ÛQe{, ${this.level + 1} Qs</div>`;
this.container.appendChild(div);
setTimeout(() => {
  if (div.parentNode) div.remove();
  this.level++;
  const cfg = this.levelConfigs[this.level - 1];
  this.speed = cfg.speed; this.targetScore = cfg.target;
  this.snake = [{ x: 10, y: 10 }];
  this.dir = { x: 1, y: 0 }; this.nextDir = { x: 1, y: 0 };
  this.spawnFood();
  this.showLevelIntro(() => {
```

=====

vÊfz\ n8b T -Æ^sSð|û-ß V3.2

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{, 29 ~u

```
    this.updateStats();
    this.loop = setInterval(() => this.tick(), this.speed);
  });
}, 1800);
},
draw() {
  const c = this.ctx, t = this.tileSize;
  c.clearRect(0, 0, this.canvas.width, this.canvas.height);
  c.fillStyle = '#ff6b6b';
  c.fillRect(this.food.x * t + 2, this.food.y * t + 2, t - 4, t - 4);
  this.snake.forEach((s, i) => {
    c.fillStyle = i === 0 ? '#a29bfe' : '#6c5ce7';
    c.fillRect(s.x * t + 1, s.y * t + 1, t - 2, t - 2);
  });
},
updateStats() {
  if (this.statsCb) this.statsCb({ 'QsSa': `${this.level}/${this.maxLevel}`
, '_R ': `${this.score}/${this.targetScore}`, '••^|': this.snake.length });
},
gameOver(won = false) {
  this.running = false; clearInterval(this.loop);
  document.removeEventListener('keydown', this._handleKey);
  const div = document.createElement('div');
  div.className = 'game-over-overlay';
  div.innerHTML = won
```

vÊfz\ n8b T -Æ^sSð|û~ß V3.2 {, 36 ~u

```
? `<h3>0<ß% `mUæ• Qsÿ </h3><p>• •ÇN†Qh•è ${this.maxLevel} Qsÿ g ~Ê_-R : ${this.score}</p><button class="btn" id="replayBtn">Q•geN \@</button>`
: `<h3>n8b ~0g_ÿ </h3><p>{, ${this.level}/${this.maxLevel} Qs | _-R : ${this.s
core}/${this.targetScore} | ••^|: ${this.snake.length}</p><button class="btn"
id="replayBtn">Q•geN \@</button>`;
this.container.style.position = 'relative';
this.container.appendChild(div);
document.getElementById('replayBtn').addEventListener('click', () => {
this.destroy(); this.init(this.container, this.statsCb); });
},
destroy() { this.running = false; clearInterval(this.loop); document.remo
veEventListener('keydown', this._handleKey); }
};
```

===== js/games/2048.js =====

```
/**
 * 2048 Game
 * @author ch2077
 * @version 1.5.3
 * @date 2025-03-08 OîY nÑR`Y1eH•î~~
 * @date 2025-05-02 XŽR Ÿ h bÖbye/c (hL-bzïN_€ybÖN†)
 *
 * ~İQx2048ep[WT ^vÿ 4x4•Qh<
 * nÑR`{-lÖÿ kİ^L/R rîzÊslideÿ QHnä0Q•T ^vv0•»v0{Iÿ g+\>^e0
 * yÜR`ziu(touch start/endwPh jîR$e-e¹T ÿ - P<20px••QM<i%æ
 * desktopziR N†mousedown/mouseupbÖbyÿ - P<15px
```

vÊfz\ n8b T -Æ^sSð|û~ß V3.2 {, 30 ~u

```
*/
const Game2048 = {
  grid: [], // 4x4ep[Wwé-5
  score: 0, // _SRM_-R ÿ T ^veö}/R ÿ
  container: null,
  statsCb: null,
  boardEl: null,
  touchStart: null,

  init(container, statsCb) {
    this.container = container;
    this.statsCb = statsCb;
    this.score = 0;
    this.touchStart = null;
```

```
this.grid = Array.from({ length: 4 }, () => Array(4).fill(0));
this.render();
this.spawn();
this.spawn();
this.updateBoard();
this.updateStats();
document.addEventListener('keydown', this._handleKey);
this.boardEl.addEventListener('touchstart', this._touchStart, {passive:
false});
this.boardEl.addEventListener('touchend', this._touchEnd, {passive:fals
e});
this.boardEl.addEventListener('mousedown', this._mouseDown);
this.boardEl.addEventListener('mouseup', this._mouseUp);
},

render() {
  this.container.innerHTML = '<div class="game-2048"><div class="board-20
48"></div></div>';
  this.boardEl = this.container.querySelector('.board-2048');
  for (let i = 0; i < 16; i++) {
    const cell = document.createElement('div');
    cell.className = 'cell-2048';
    this.boardEl.appendChild(cell);
  }
},

spawn() {
  const empty = [];
  for (let r = 0; r < 4; r++)
    for (let c = 0; c < 4; c++)
      if (this.grid[r][c] === 0) empty.push({ r, c });
  if (empty.length === 0) return;
  const { r, c } = empty[Math.floor(Math.random() * empty.length)];
  this.grid[r][c] = Math.random() < 0.9 ? 2 : 4;
},

updateBoard() {
  const cells = this.boardEl.querySelectorAll('.cell-2048');
  cells.forEach((cell, i) => {
```

```
const r = Math.floor(i / 4);
const c = i % 4;
const v = this.grid[r][c];
```

```
cell.textContent = v || '';
cell.className = 'cell-2048' + (v ? ` tile-${v}` : '');
});
},

updateStats() {
  if (this.statsCb) this.statsCb({ '_R ': this.score });
},

doMove(dir) {
  let moved = false;
  const old = this.grid.map(r => [...r]);
  if (dir === 'up') moved = this.moveUp();
  else if (dir === 'down') moved = this.moveDown();
  else if (dir === 'left') moved = this.moveLeft();
  else if (dir === 'right') moved = this.moveRight();
  else return;
  if (moved) {
    this.spawn();
    this.updateBoard();
    this.updateStats();
    if (this.checkWin()) {
      this.showResult('0<ß% 0`•bN†ÿ •%R0 2048ÿ ');
    } else if (this.checkLose()) {
      this.showResult('n8b ~0g_ÿ lig SïyûR`v„N†');
    }
  }
},

_handleKey(e) {
  const map = { ArrowUp:'up', ArrowDown:'down', ArrowLeft:'left', ArrowRight:'right' };
  if (map[e.key]) { e.preventDefault(); Game2048.doMove(map[e.key]); }
},

_touchStart(e) { e.preventDefault(); Game2048.touchStart = { x: e.touches[0].clientX, y: e.touches[0].clientY }; },
_touchEnd(e) {
  e.preventDefault();
  if (!Game2048.touchStart) return;
  const dx = e.changedTouches[0].clientX - Game2048.touchStart.x;
  const dy = e.changedTouches[0].clientY - Game2048.touchStart.y;
  Game2048.touchStart = null;
  if (Math.abs(dx) < 20 && Math.abs(dy) < 20) return;
  if (Math.abs(dx) > Math.abs(dy)) {
    Game2048.doMove(dx > 0 ? 'right' : 'left');
  } else {
    Game2048.doMove(dy > 0 ? 'down' : 'up');
  }
}
```

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vÊfz\ n8b T -Æ^sSð|û~ß V3.2

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{, 39 ~u

}

=====

vÊfz\ n8b T -Æ^sSð|û~ß V3.2

=====

{, 32 ~u

},

```
_mousedown(e) { Game2048.touchStart = { x: e.clientX, y: e.clientY }; },
_mouseup(e) {
  if (!Game2048.touchStart) return;
  const dx = e.clientX - Game2048.touchStart.x;
  const dy = e.clientY - Game2048.touchStart.y;
  Game2048.touchStart = null;
  if (Math.abs(dx) < 15 && Math.abs(dy) < 15) return;
  if (Math.abs(dx) > Math.abs(dy)) {
```